



VIDYARTHI VIGYAN MANTHAN 2026-27

SYLLABUS (CLASS-VII)

CLASS VII

PHYSICS

- 1. Motion:** Appreciation of the idea of time and the need to measure it. Measurement of time using periodic events. Idea of speed of moving objects – slow and fast motion along a straight line. Observing and analysing the motion (slow or fast) of common objects on land, in air, water and space. Measuring the distance covered by objects moving on a road in a given time and calculating their speeds. Plotting distance vs. time graphs for uniform motion. Measuring the time taken by moving objects to cover a given distance and calculating their speeds. Constancy of the time period of a pendulum.
- 2. Light:** Rectilinear propagation of light, Reflection, certain surfaces reflect light. Real and virtual images. White light is composed of many colours. Observation of the source of light through a straight tube, a bent tube. Observing the reflection of light on a wall or a white paper screen. Open-ended activities allow children to explore images made by different objects and record observations. Focused discussions on real and virtual images. Making the colour disc and rotating it.
- 3. Electricity:** Electric circuit symbols for different elements of a circuit. The heating effect of current. Principle of fuse. A current-carrying wire has an effect on a magnet. A current-carrying coil behaves like a magnet. The working of an electric bell. Drawing circuit diagrams. Activities to show the heating effect of electric current. Making a fuse. Activity to show that a current-carrying wire has an effect on a magnet. Making a simple electromagnet. Identifying situations in daily life where electromagnets are used. Demonstration of the working of an electric bell.
- 4. Rain, Thunder and Lightning:** High-speed winds and heavy rainfall have disastrous consequences for human and other life. Making wind speed and wind direction indicators. Activity to show “lift” due to moving air. Discussion on the effects of storms and possible safety measures

CHEMISTRY

- Exploring substances: Acidic, Basic and Neutral
- The World of Metals & Non - Metals
- Changes around us: Physical & Chemical
- Heat Transfer in Nature

BIOLOGY

- Adolescence
- Life processes in animals
- Life processes in plants

MATHEMATICS



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1. **Integers:** Brahmagupta rules on zeros, Number game, Integer Magic Square
2. **Fractions and Decimals:** Types of Fraction, Expressing a fraction into unit fraction using Mahavira's rule, Operations on Fraction and Decimals
3. **Rational Numbers:** Definition, Properties of Rational Numbers
4. **Exponents and Powers:** Integral Exponent, Finding unit digits of any number with powers up to 1000.
5. **Algebraic Expressions:** Multiplication and division of Polynomials
6. **Linear Equations in One Variable:** Solving linear equations, problems based on it.
7. **Arithmetic:** Percentages, Profit-Loss, Simple Interest, Speed, Time and Distance
8. **Geometry:** Lines and Angles, Triangles and Properties, Congruence of Triangles
9. **Mensuration:** Perimeter and Area (2D Geometry), Mensuration (Volume and Surface Area)
10. **Data Handling:** Mean, Median, Mode, Graphs- Bar, Multiple bar, histogram
11. **Visualising Solid Shapes:** Archimedean shapes, Verifying Euler formula

LOGIC AND REASONING

Pattern Recognition: Number Series, Alphabet Series, Image Series, Odd One Out, Analogy, Figure Matrix & Matrix Reasoning, Pattern-based Puzzles, Coding-Decoding, Logical Sequence of Words.

Analytical Reasoning: Blood Relations, Direction Sense, Order & Ranking, Seating Arrangement, Grid Puzzles, Games & Tournaments, Scheduling Problems, Clocks & Calendar, Sequencing & Ordering Problems.

Visual Reasoning: Mirror Images, Water Images, Rotation, Paper Folding, Paper Cutting, Figure Counting, Embedded Figures, Shape Construction, Spatial Visualization, 3D visualization, Cubes, Dice & Cube Nets, Best Fit, Symmetry.

Mathematical Reasoning: Mathematical Operations, Number Puzzles, Cryptarithmic (Alphametics) puzzles, Binary Logic, Logical Inequalities, Logical Venn Diagrams, Word Problems.

Critical Thinking: Data Sufficiency, Decision Making, Input Output series, Critical Path analysis.

LIFE STORY OF INDIAN SCIENTIST(S)



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Anna Mani : The Weather Woman of India (All Chapters)

INDIAN CONTRIBUTIONS TO SCIENCE

Indian Contributions to Science : Volume-I